

Spatially Explicit Rock–TBM Interaction Graph Sequences for Response-Consistent Interpretation of TBM Excavation Processes

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Abstract

Existing representations of rock–TBM interaction during excavation treat monitoring responses as temporal sequences, which preserve temporal dependence but lack the spatial explicitness needed to resolve where interactions occur, which machine components are implicated, and how interaction structure evolves along chainage. This study proposes a response-supervised, geometry-constrained graph sequence framework for spatially explicit representation and interpretable learning of rock–TBM interaction. The framework introduces three key innovations: (i) a heterogeneous graph-sequence representation that formalizes TSP-derived rock voxels and parameterized TBM surface nodes as spatial entities with explicit candidate interaction relations; (ii) a geometry-constrained edge construction strategy based on spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints, which restricts learning to spatially plausible rock–machine relations rather than unconstrained connectivity; and (iii) a diagnostic relevance aggregation procedure that summarises raw rock–machine edge scores at surface-node, component, and chainage levels to support response-consistent spatial diagnostics. Rather than relying on sparse jamming labels or manually calibrated contact forces, the framework uses continuously recorded monitoring responses to supervise the relative importance of geometry-screened candidate relations, enabling interpretable spatial learning under incomplete observability. The evaluation uses two tunnel cases: BSLL DyK1017+205 and SJLS Dyk1252+411. BSLL is used for one-step, multi-step, and interpretation analysis, while SJLS provides an external TSP case for testing prediction consistency and geometry ablation. The framework remains competitive with temporal baselines, and SJLS is consistent with a small case-specific geometry-related MAE difference, but the experiments do not establish a broad forecasting improvement. The more consistent evidence is interpretive: checkpoint-based diagnostics show positively clustered diagnostic relevance in all evaluated runs, relevance that is not simply explained by candidate-edge degree, and case-specific association with local TSP velocity and monitoring responses. These patterns are reported as response-consistent diagnostics, not as proof of actual contact conditions or geological causality. By unifying geological sensing, machine geometry, monitoring response, and diagnostic relevance within a constrained graph sequence, the framework offers a spatially explicit approach to excavation interaction analysis that complements existing temporal prediction methods.

Keywords: spatial representation; heterogeneous graph sequence; rock–machine interaction; geometry-constrained edge construction; response supervision; interpretable geospatial learning; TBM excavation; diagnostic relevance

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1 Introduction

Tunnel boring machines (TBMs) are widely used in long and deep tunnel construction, where excavation responses emerge from the coupled effects of geological conditions, machine geometry, and operational process. Under fractured, water-rich, high-stress, or locally deformable ground conditions, the interaction state between surrounding rock and the TBM may change rapidly and manifest through abnormal thrust, torque, penetration, advance rate, or shield-related responses. These monitoring variables are informative external expressions of rock–TBM interaction, but the geological body and the cutterhead or shield form a spatial relation that evolves as excavation advances. Monitoring curves alone provide limited evidence about where the interaction occurs, which machine components are implicated, and how local interaction relations evolve along chainage.

Most existing methods process TSP-derived geological indicators and TBM parameters as tabular variables or time-series variables. Such representations can support trend prediction, but they cannot answer which rock spatial unit forms a candidate interaction relation with which machine component at a given excavation step [1, 5, 6, 19]. Sequence models preserve temporal dependence but treat excavation as an ordered signal rather than a spatially structured interaction process, so rock location, machine component structure, and local rock–machine candidate relations are not retained within a unified representation.

Rather than pursuing a more complex prediction model, this study constructs a chainage-referenced rock–TBM interaction graph sequence that organises rock voxels, TBM surface samples, and operational response sequences into a computable spatio-temporal graph. TSP-derived geological attributes are formalised as rock voxel nodes, while the cutterhead and shield are discretised as parameterised TBM surface nodes. Candidate rock–machine edges are screened through spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints. Instead of interpreting edge weights as contact-force measurements or relying on sparse jamming labels, the framework uses continuously recorded monitoring responses to supervise the relative importance of geometry-screened candidate relations, and aggregates the learned relevance at edge, surface-node, component, and chainage levels to generate response-consistent diagnostic views.

The main contributions are threefold. First, the study proposes a chainage-referenced rock–TBM spatial representation that formalises heterogeneous spatial entities—TSP voxels, TBM surface components, and chainage-indexed monitoring responses—into a unified entity model preserving geometric and topological properties. Second, it proposes a geometry-constrained rock–machine interaction graph sequence in which candidate edges are restricted by active-zone membership, distance, surface-normal compatibility, and excavation state, ensuring that learned relations remain geometrically plausible rather than unconstrained. Third, it uses response-supervised prediction to examine the response consistency of the learned graph relations and outputs edge-, component-, and chainage-level relevance diagnostics, with clear interpretation limits, as spatially organised diagnostics rather than contact-force or risk estimates.

2 Related work

2.1 TBM response prediction and jamming-related analysis

Existing TBM prediction and warning studies have shown that tabular models and sequence models can be effective for forecasting penetration, thrust, torque, or jamming-related responses [1, 5, 19]. Sequence models are effective for learning temporal dependence in monitoring records [6], but they usually treat excavation as an ordered signal sequence rather than as a spatially structured interaction process. As a consequence, existing studies usually represent geological and machine variables as feature vectors or sequences, rather than explicit spatial relations

between geological entities and TBM components. This limitation matters when the objective is not only response prediction, but also spatially consistent interpretation of interaction dynamics.

2.2 Spatial representation of geological conditions in tunnel excavation

Tunnel seismic prediction (TSP) and advance geological prediction provide spatially distributed descriptions of the geological field ahead of or around excavation [7]. TSP-derived geological information can be transformed into chainage-referenced rock voxels that preserve spatial support and local geological context. Voxelised geological models and three-dimensional geological representation have been used to describe subsurface heterogeneity along tunnel alignments. However, this representation still needs to be linked to machine geometry: a geological voxel field alone does not specify which rock units are candidate interaction partners for which TBM surface components at a given excavation step. Bridging geological spatial representation and machine surface geometry is therefore a prerequisite for spatially explicit rock-machine interaction modelling.

2.3 Graph-based spatiotemporal prediction and spatial interaction modelling

Graph learning offers a natural alternative because it is well suited to irregular objects and relations. In GIScience, graph concepts have long been used to formalise spatial contiguity and spatial interaction [16], and recent IJGIS and IJDE studies show how graph neural networks can represent spatial units, neighbourhood structure, and urban network morphology [3, 9]. Beyond geography-specific applications, graph-based models have been used to encode object interaction, state propagation, and structured spatial systems in domains such as physics simulation [10, 11], urban flow prediction [4, 21], and spatio-temporal forecasting [8, 18]. Heterogeneous graph networks further enable the representation of multi-type entities and relations [12, 15, 20]. In graph-based spatiotemporal prediction, the graph is not merely a computational backbone but an explicit encoding of spatial entities and their relations; the prediction problem is defined on a spatial graph together with a feature matrix, after which graph construction and model design follow. Yet TBM excavation is not a generic graph-learning problem in which edges can be learned freely from black-box correlation. Candidate rock-TBM relations must remain consistent with machine geometry, excavation state, and local active zones. In this setting, graph learning is most meaningful when it operates on geometrically plausible candidate relations rather than on unconstrained connectivity.

3 Methodology

This section formalizes the proposed framework as a geometry-constrained heterogeneous graph sequence supervised by continuously recorded TBM monitoring response. The method represents excavation as an evolving system of spatial entities—rock units and TBM surface components—and their candidate interaction relations, rather than as a monitoring-only signal sequence. By formalizing spatial entities, constraining candidate relations by geometric compatibility, and aggregating learned relevance into component- and chainage-resolved diagnostics, the framework addresses the geographic-information representation problem identified in the Introduction: how to organize heterogeneous spatial entities into a representation that preserves spatial explicitness, relation plausibility, and interpretability.

3.1 Chainage-referenced rock-TBM representation

Let excavation advance be discretized into chainage-indexed steps $t = 1, 2, \dots, T$. For each step, the framework constructs a graph snapshot G_t and then models the ordered graph sequence $\{G_t\}_{t=1}^T$. The framework is organised into three coupled levels: chainage-referenced rock-TBM

representation, geometry-constrained relation graph construction, and response-supervised diagnostic learning (Figure 1). Given the most recent K excavation steps, the model uses graph snapshots together with synchronized TBM monitoring vectors to predict a future response at step $t + h$:

$$\hat{r}_{t+h} = f_{\Theta}(G_{t-K+1:t}, u_{t-K+1:t}),$$

where u_t denotes the monitoring vector at step t , r_{t+h} denotes the target future response, K is the historical window length, and h is the forecasting horizon. In this study, the learned rock-machine raw edge scores and derived diagnostic relevance are interpreted as response-consistent model diagnostics under the forecasting task, not as true contact force, contact pressure, jamming probability, or any direct engineering risk quantity.

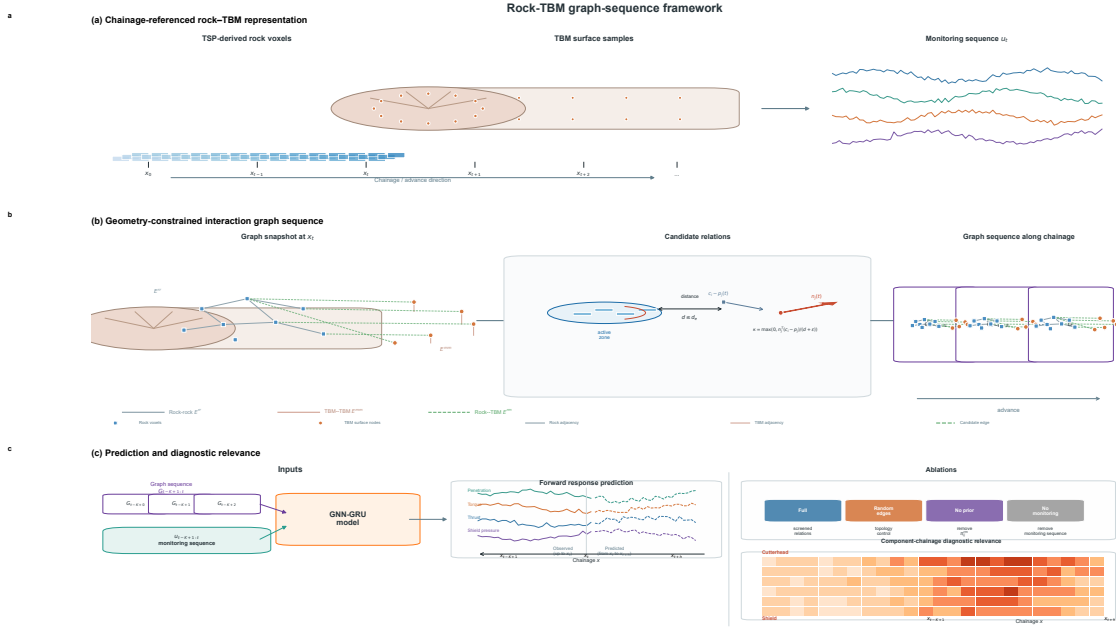


Figure 1: Overview of the proposed rock-TBM interaction graph-sequence framework. The figure summarises three methodological levels: chainage-referenced rock-TBM representation, geometry-constrained interaction graph-sequence construction, and response-supervised prediction with diagnostic relevance views. The relevance outputs are model-derived diagnostics and are not interpreted as contact force, contact pressure, or calibrated jamming risk.

The monitoring input vector can include

$$u_t = [\text{AdvanceRate}_t, \text{RPM}_t, \text{Torque}_t, \text{Thrust}_t, \text{Penetration}_t, \text{ShieldPressure}_t],$$

while the prediction target can be defined as a multivariate future response vector

$$r_{t+h} = [\text{AdvanceRate}_{t+h}, \text{Torque}_{t+h}, \text{Thrust}_{t+h}, \text{Penetration}_{t+h}, \text{ShieldPressure}_{t+h}].$$

Because TSP coordinates and TBM shield mileage are usually recorded in different local reference systems, the method also requires a local offset mapping. A simple alignment is to define

$$x_{\text{local}}(t) = \text{ShieldMileage}(t) - \text{ShieldMileage}_{\text{start}},$$

and then map $x_{\text{local}}(t)$ into the TSP coordinate range used for voxelized geological representation.

Geological and machine inputs. The geological input is a TSP-derived voxel field $D_{\text{geo}} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ denotes the spatial coordinate of rock voxel i , and g_i denotes its

geological attribute vector. Depending on the available inversion and interpretation products, g_i may include seismic velocity, elastic indicators, ratio features, or other derived geological descriptors.

The TBM input is represented as a parameterized surface model $M_{\text{TBM}} = \{p_j(0), n_j, \rho_j\}$, where $p_j(0)$ is the initial coordinate of TBM surface node j , n_j is the local surface normal, and ρ_j is a machine component label such as cutterhead or shield. The excavation process is further described by advance direction d , step length Δx , step-dependent active zone Ω_t , candidate-edge distance threshold τ_{edge} , and normal-compatibility threshold η_{min} . When a detailed three-dimensional TBM model is unavailable, the surface can be approximated by a parameterized geometry in which the cutterhead is represented as a disc and the shield as cylindrical surface components. In the current formulation, the surface is partitioned into cutterhead, front shield, middle shield, and tail shield, and nodes are sampled at fixed radial, circumferential, and axial resolutions.

3.2 Geometry-constrained rock-machine relation graph sequence

Graph-sequence construction. At each excavation step t , a graph snapshot is defined as

$$G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}),$$

where V_t^r and V_t^m are rock and TBM node sets, respectively, E_t^{rr} denotes rock-rock adjacency, E_t^{mm} denotes TBM-surface adjacency, and E_t^{rm} denotes candidate rock-machine interaction edges.

Rock node features are defined as

$$x_i^r(t) = [c_i, g_i, s_i(t)],$$

where $s_i(t)$ records whether a voxel remains active and eligible for interaction at step t . TBM node features are defined as

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j].$$

With constant advance step length, TBM surface nodes are updated along the advance direction by

$$p_j(t+1) = p_j(t) + \Delta x d.$$

For intra-rock connectivity, E_t^{rr} is constructed from voxel adjacency, for example a 26-neighbourhood relation. For TBM-surface connectivity, E_t^{mm} is constructed from local surface neighbourhood relations inherited from the discretized machine geometry. The key design choice concerns the candidate interaction edges E_t^{rm} , which are not fully connected but screened by geometric plausibility. This constrained construction contrasts with unconstrained graph learning [17, 22] and reflects the domain requirement that candidate rock-machine relations must be geometrically plausible.

Geometry-constrained candidate interaction edges. A rock voxel i and TBM surface node j are linked at step t only if the candidate relation satisfies a geometry-constrained rule:

$$e_{ij}^{rm}(t) = 1$$

if

$$\begin{cases} d_{ij}(t) \leq \tau_{\text{edge}}, \\ \kappa_{ij}(t) \geq \eta_{\text{min}}, \\ c_i \in \Omega_t, \\ s_i(t) = 1, \end{cases}$$

and the edge is absent otherwise. Here,

$$d_{ij}(t) = \|c_i - p_j(t)\|$$

is the rock-to-surface distance, and

$$\kappa_{ij}(t) = \max \left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon} \right)$$

measures signed-normal compatibility between the TBM surface orientation and the rock-voxel direction. The active-zone condition restricts candidate relations to the geological region currently relevant to excavation, while the state variable $s_i(t)$ prevents already excavated or inactive voxels from repeatedly contributing to later interaction snapshots.

This design intentionally limits graph learning to a screened candidate set. The edge relevance learned later in the model is therefore interpreted as the relative importance of geometrically plausible candidate interactions, rather than as a direct estimate of contact force or stress.

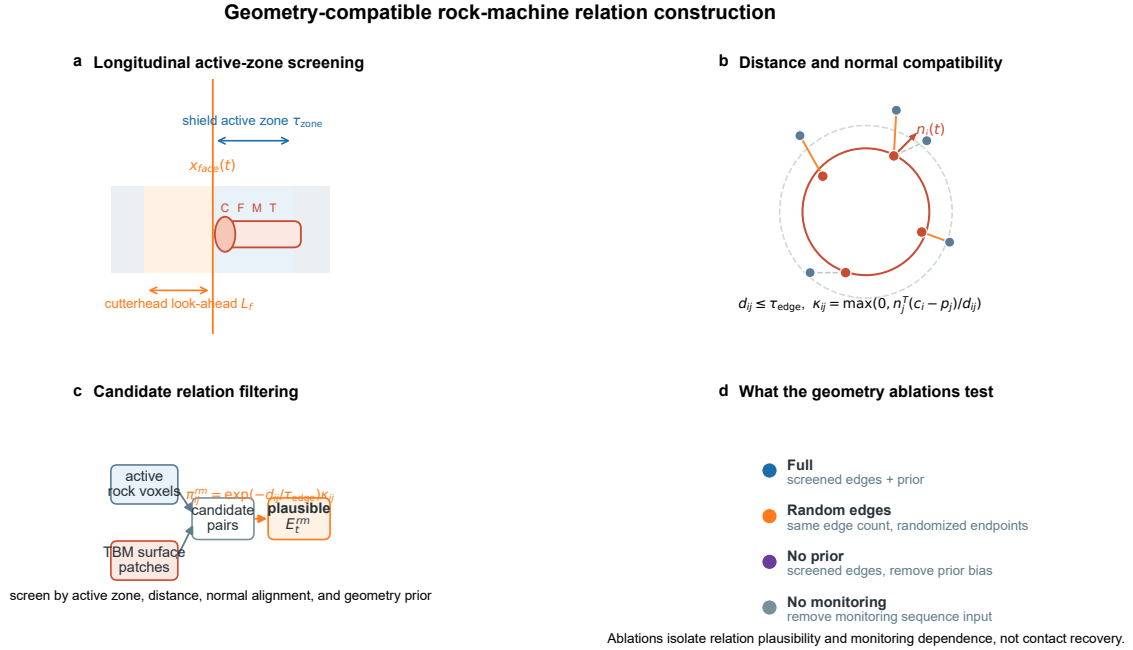


Figure 2: Geometry-compatible rock-machine relation construction. Candidate edges are restricted by active-zone membership, distance, surface-normal compatibility, and the geometric prior π_{ij}^{rm} . The ablation variants test relation plausibility and prior guidance; they are not interpreted as contact recovery experiments.

Edge features and geometric prior. Each candidate rock-machine edge is further associated with an edge-feature vector

$$a_{ij}^{rm} = [d_{ij}/\tau_{\text{edge}}, \kappa_{ij}, (c_i - p_j)/\tau_{\text{edge}}, g_i, \text{onehot}(\rho_j)],$$

which combines three kinds of information: geometric compatibility, local geological context, and machine-component semantics. In addition, a simple geometric prior is defined as

$$\pi_{ij}^{rm} = \exp(-d_{ij}/\tau_{\text{edge}}) \kappa_{ij}.$$

This prior increases when the rock voxel is closer to the TBM surface and more consistent with the local outward normal. It is used only as geometric guidance for learning and should not be interpreted as contact force, contact pressure, or a direct risk indicator. The full graph-sequence construction procedure is summarised in Algorithm 1.

Algorithm 1 Geometry-constrained graph-sequence construction

Require: Rock voxel field D_{geo} , TBM surface M_{TBM} , thresholds τ_{edge} , η_{min} , active-zone Ω_t

- 1: **for** each excavation step $t = 1, \dots, T$ **do**
- 2: Update TBM surface positions: $p_j(t) \leftarrow p_j(t-1) + \Delta x d$
- 3: Build rock–rock edges E_t^{rr} from 26-neighbourhood
- 4: Build TBM–TBM edges E_t^{mm} from surface adjacency
- 5: Build candidate rock–machine edges:

$$E_t^{rm} \leftarrow \{(i, j) : d_{ij}(t) \leq \tau_{\text{edge}}, \kappa_{ij}(t) \geq \eta_{\text{min}}, c_i \in \Omega_t, s_i(t) = 1\}$$

- 6: Compute edge features a_{ij}^{rm} and geometric prior π_{ij}^{rm}
 - 7: Assemble graph snapshot $G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm})$
 - 8: **end for**
 - 9: **return** Graph sequence $\{G_t\}_{t=1}^T$
-

3.3 Response-supervised graph-sequence learning and diagnostic relevance

Response-supervised interaction learning. After graph construction, each snapshot is encoded by a graph encoder to produce an interaction-aware latent representation. Message passing integrates geological attributes, machine-surface context, and candidate interaction structure across the heterogeneous graph. The resulting stepwise graph embeddings are then passed to a temporal encoder, which models graph-state evolution along chainage and links the sequence of graph states to continuously recorded TBM monitoring response.

For each graph snapshot, the graph encoder computes raw (pre-softmax) edge scores $s_{ij}^{rm}(t)$ for each candidate rock–machine edge, which are then normalised over all source rock nodes incident to each destination TBM node j to obtain attention weights:

$$\alpha_{ij}^{rm}(t) = \frac{\exp(s_{ij}^{rm}(t))}{\sum_{i': (i', j) \in E_t^{rm}} \exp(s_{i'j}^{rm}(t))}.$$

These weights are used in message passing [13, 14]. The encoder also produces a graph-level representation z_t . The graph-level representation is concatenated with the synchronized monitoring vector u_t and passed to a gated recurrent unit (GRU) [2], which models graph-sequence evolution across consecutive excavation steps.

The supervision signal is provided by monitoring response variables recorded during excavation, rather than by sparse binary jamming labels or manually assigned edge weights. Under this design, the model learns which screened candidate rock–machine relations are most relevant for explaining or anticipating response variation. The resulting raw scores $s_{ij}^{rm}(t)$ are therefore response-consistent relevance indicators under geometric screening. The training procedure is summarised in Algorithm 2.

Diagnostic relevance aggregation. To recover spatial interpretability, the learned relevance associated with candidate rock–machine relations is aggregated into surface-node, component, and chainage summaries.

Notation. We distinguish three quantities in the interpretation pipeline:

- *Raw edge score* $s_{ij}^{rm}(t)$: the pre-softmax attention score computed by the graph encoder for candidate edge (i, j) at step t . This quantity preserves absolute magnitude differences across edges and is used for all spatial interpretation outputs.

Graph-sequence network for response-supervised spatial learning

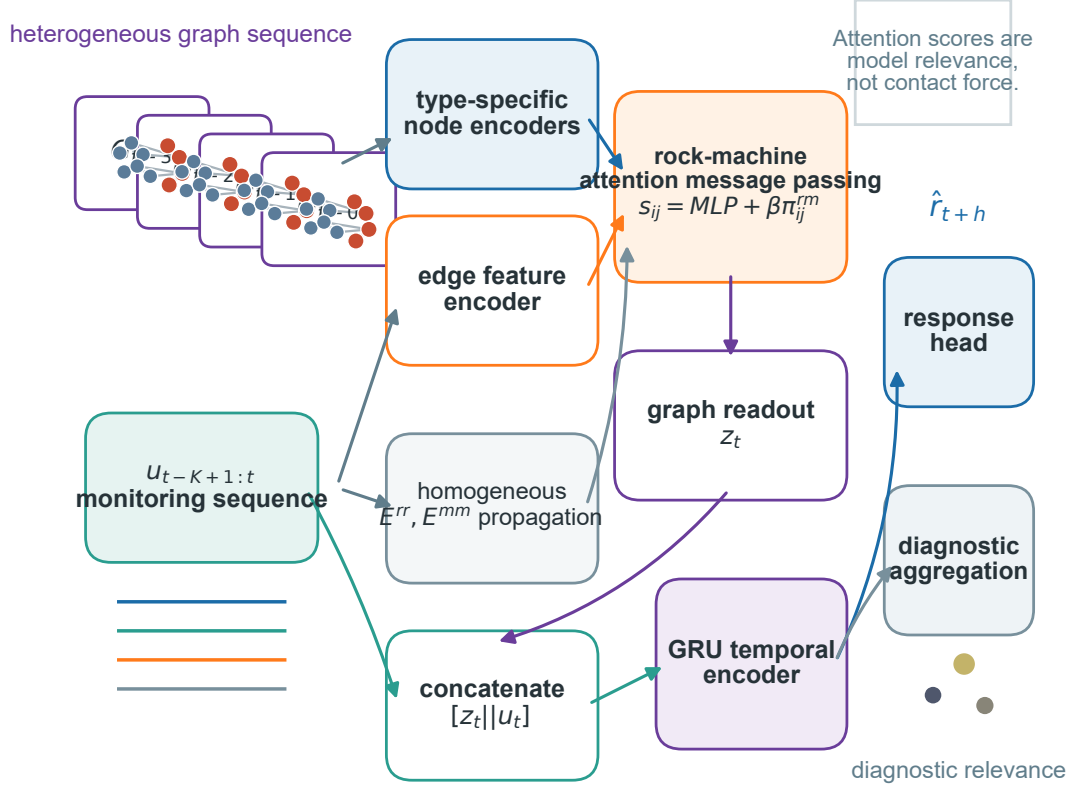


Figure 3: Graph-sequence network architecture. Each graph snapshot is encoded by type-specific node encoders, an edge-feature encoder, geometry-guided rock-machine attention message passing, and graph readout. The graph embedding is concatenated with the monitoring sequence and passed through a GRU temporal encoder to predict future response. Raw edge scores are aggregated as model diagnostic relevance, not contact-force measurements.

Algorithm 2 Response-supervised interaction learning

Require: Graph sequence $\{G_t\}$, monitoring vectors $\{u_t\}$, window K , horizon h

- 1: **for** each training step **do**
 - 2: **for** each sample window $t - K + 1$ to t **do**
 - 3: Encode each G_s by heterogeneous GNN: $z_s, \{s_{ij}^{rm}(s)\} \leftarrow \text{GNN}(G_s)$
 - 4: Concatenate graph and monitoring: $h_s = [z_s; u_s]$
 - 5: **end for**
 - 6: Model temporal evolution: $\{h_s\} \leftarrow \text{GRU}(\{h_s\})$
 - 7: Predict: $\hat{r}_{t+h} = \text{FC}(h_T)$
 - 8: Compute loss: $\mathcal{L} = \text{Huber}(\hat{r}_{t+h}, r_{t+h})$
 - 9: Update parameters Θ by back-propagation
 - 10: **end for**
 - 11: **return** Trained model f_Θ , raw edge scores $\{s_{ij}^{rm}(t)\}$
-

- *Normalised attention weight $\alpha_{ij}^{rm}(t)$* : the post-softmax weight used in message passing, obtained by normalising raw scores over all source rock nodes incident to destination TBM node j :

$$\alpha_{ij}^{rm}(t) = \frac{\exp(s_{ij}^{rm}(t))}{\sum_{i':(i',j) \in E_t^{rm}} \exp(s_{i'j}^{rm}(t))}.$$

Because $\sum_{i:(i,j) \in E_t^{rm}} \alpha_{ij}^{rm}(t) = 1$ for each TBM node j , aggregating α_{ij}^{rm} per node yields a constant and is therefore unsuitable for spatial interpretation.

- *Surface-node diagnostic relevance $C_j(t)$* : the degree-normalised aggregated non-negative-transformed raw score incident to TBM surface node j , defined as

$$C_j(t) = \frac{1}{d_j(t) + \epsilon} \sum_{i:(i,j) \in E_t^{rm}} \text{softplus}(s_{ij}^{rm}(t)),$$

where $d_j(t) = |\{i : (i,j) \in E_t^{rm}\}|$ is the degree of TBM node j in the rock-machine sub-graph at step t , $\epsilon = 10^{-8}$ avoids division by zero, and $\text{softplus}(x) = \log(1 + e^x)$ is a smooth non-negative transform. Softplus transforms raw pre-softmax scores into non-negative relevance summaries, while degree normalisation prevents surface nodes with more incident candidate edges from being trivially assigned higher relevance. No further per-chainage standardisation is applied to C_j for the quantitative spatial consistency indicators.

Aggregation. $C_j(t)$ is summarised over TBM components and tracked along excavation chainage to form component- and chainage-resolved diagnostic relevance views. These outputs indicate where the response-supervised model places diagnostic emphasis within the screened candidate relation set, while remaining bounded by the monitored response signal used for supervision.

4 Case studies and experimental design

4.1 Case roles and data sources

The experiments use two real tunnel cases with complementary evidential roles. The first is the BSLL tunnel case starting from DyK1017+205. It provides the main graph-sequence construction example, the one-step response prediction run, and the three-step advance-prediction sensitivity run. The second is the SJLS tunnel case starting from Dyk1252+411. It uses an external TSP-derived velocity field and real mileage-aligned TBM monitoring labels, and is used to test whether the same spatial representation remains meaningful in a second tunnel setting.

The two cases are not treated as sequential data releases or replacement datasets. BSLL is used to examine one-step and multi-step settings, while SJLS provides an external TSP contrast. BSLL is used mainly to illustrate multi-step prediction and spatial interpretation under a compact test interval. SJLS is used mainly for prediction consistency and geometry-ablation evidence under an external TSP case.

For each excavation step, the monitoring vector contains advance rate, cutterhead RPM, torque, thrust, penetration, and shield pressure. The prediction target is the future multivariate response vector consisting of advance rate, torque, thrust, penetration, and shield pressure. The geological input is a TSP-derived voxel field. For BSLL, the project TSP voxel field contains P-wave velocity, S-wave velocity, density, dynamic Young’s modulus, velocity ratio, and Poisson’s ratio around the tunnel alignment. For SJLS, dense external P-wave and S-wave velocity profiles at Dyk1252+411 are preprocessed into the same project TSP schema and paired with mileage-aligned monitoring labels.

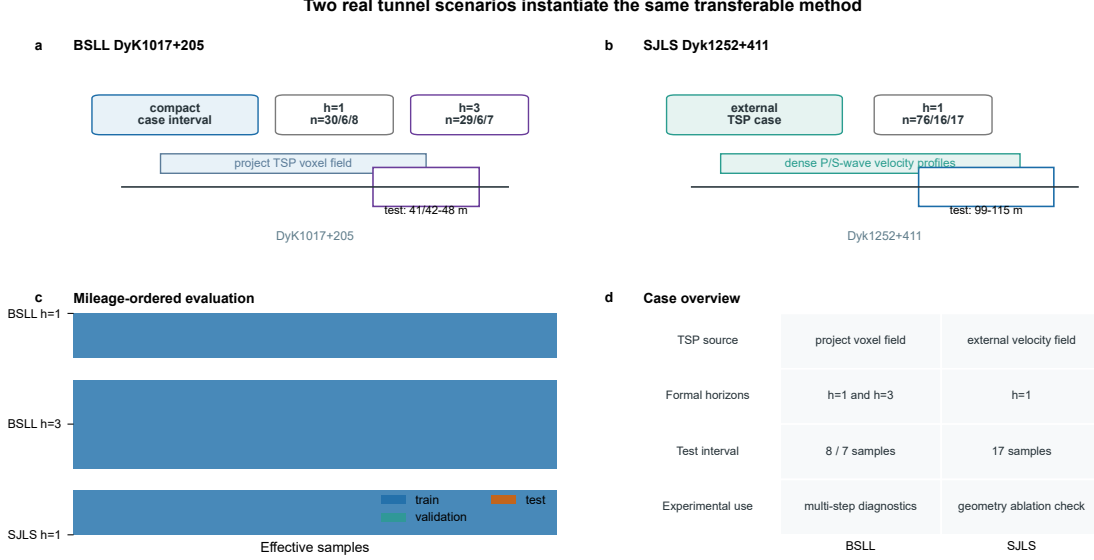


Figure 4: Real tunnel scenarios used to instantiate the common method. BSLL DyK1017+205 provides compact one-step and three-step case runs for prediction checks, multi-step behaviour, and diagnostic interpretation. SJLS Dyk1252+411 provides an external TSP case for prediction consistency and geometry-ablation evidence.

4.2 Excavation-step samples and graph settings

All experiments use mileage-ordered partitioning to preserve forward prediction. For a historical window of $K = 5$, each sample uses the preceding monitoring and graph snapshots as input and predicts a future response at horizon h . The formal case runs are summarised in Table 1.

Table 1: Formal case runs used in the experiments. Effective samples are sequence windows after applying the history window and prediction horizon.

Case	Tunnel	Start chainage	Horizon	Train/Val./Test samples
BSLL h=1	BSLL	DyK1017+205	1	30 / 6 / 8
BSLL h=3	BSLL	DyK1017+205	3	29 / 6 / 7
SJLS h=1	SJLS	Dyk1252+411	1	76 / 16 / 17

The BSLL one-step run contains 30 training samples, 6 validation samples, and 8 test samples (target chainages 41–48 m). The BSLL three-step run contains 29 training samples, 6 validation samples, and 7 test samples (target chainages 42–48 m). The SJLS run contains 76 training samples, 16 validation samples, and 17 test samples (target chainages 99–115 m).

The graph construction uses $\tau_{\text{edge}} = 2.0$ m, $\eta_{\text{min}} = 0.3$, and $\tau_{\text{zone}} = 5.0$ m. The TBM surface is discretised into 1352 nodes (cutterhead disc + front, middle, and tail shield cylinders) at a surface resolution of 0.5 m. The TSP voxel field yields approximately 3500 active-zone rock nodes per excavation step. The historical window length is $K = 5$. The BSLL one-step and SJLS cases use $h = 1$, while the BSLL multi-step sensitivity case uses $h = 3$.

The TSP preprocessing follows the same voxel schema for both cases. Coordinates are converted to a local excavation coordinate by subtracting the minimum longitudinal coordinate, so the local X axis is aligned with advance direction. Each voxel stores $[V_p, V_s, E, V_p/V_s, \nu, \rho]$, and these attributes are standardised using statistics fitted on the training active-zone margin. For SJLS, the external velocity profiles are first averaged onto a 1 m X – Y profile grid, then expanded to a 1 m vertical grid over ± 10 m to match the project voxel schema. At each

excavation step, active rock nodes are selected from two component-aware regions: a cutter-head look-ahead region in front of the face and a shield annular region around the shield body. The active-zone range τ_{zone} is therefore used for node selection, whereas τ_{edge} is used later for candidate rock-machine edge screening.

All model and optimiser settings are kept fixed across the formal case runs and recorded in the experiment configuration files. Validation samples are used only for learning-rate scheduling and early stopping; all reported metrics are computed on the mileage-ordered test intervals.

4.3 Baselines and graph ablation settings

The experimental design compares the proposed framework against a persistence baseline and graph-structure ablations that are reported in the result tables.

Prediction baseline. Persistence uses the most recent observed response as the forecast. The full graph-sequence model uses the heterogeneous graph sequence with geometry-constrained candidate edges.

Graph ablations. Random-edge replaces the geometry-constrained candidate edges with randomised rock-machine edges of matched density. No-prior removes the geometric prior π_{ij}^{rm} from the edge features. No-monitoring removes the monitoring-sequence input from the predictor, retaining the graph sequence. These ablations isolate the contribution of graph topology, geometric prior guidance, monitoring-sequence input, and geometry-based candidate screening.

4.4 Evaluation and reporting principle

The formal evaluation uses a chronological mileage-ordered split. The first 70% of prediction-target samples are used for training, the next 15% for validation, and the remaining 15% for testing. This design evaluates forward generalisation to a later continuous tunnel interval and deliberately retains the geological distribution shift encountered by the advancing TBM.

An important reporting principle is stated in advance: prediction metrics are used as forward response checks, whereas relevance diagnostics are used to examine whether the learned graph relations are spatially organized. Moran’s I is computed using inverse-distance weights on TBM surface node positions within each chainage. Component CV is the coefficient of variation of mean C_j across the four TBM components (cutterhead, front, middle, tail shield). Attention-geology correlation is the Pearson and Spearman correlation between mean C_j and TSP V_p across test chainages. These two evaluation axes—prediction accuracy and spatial diagnostic quality—are reported separately and should not be conflated.

5 Results

5.1 Forward prediction check across case roles

Table 2 reports the case-level forward prediction metrics. The results support a cautious interpretation. In BSLL, the one-step graph-sequence model obtains MAE 0.4147, essentially matching Persistence (0.4146) and the random-edge variant (0.4145). The three-step BSLL run similarly remains close to Persistence (0.4617 versus 0.4618), while the random-edge variant is slightly lower in MAE (0.4609). These BSLL results do not support a claim that the graph sequence improves prediction in this case.

The SJLS case gives a more favourable but still small effect. The full model obtains MAE 0.2553, slightly lower than Persistence (0.2554), randomised edges (0.2559), and the no-prior variant (0.2570). The Pearson correlation is high ($r = 0.960$), but R^2 remains negative because the mileage-ordered test interval is a demanding extrapolation setting. Thus the SJLS result is

best read as a small response-consistency difference under the external TSP case, not as a large prediction breakthrough.

Table 2: Forward prediction results for the semantic case runs. MAE and RMSE are computed on standardised multivariate response targets. The random-edge column reports the corresponding graph-topology ablation.

Case	Test n	Full MAE	Full RMSE	R^2	Persistence MAE	Random-edge MAE
BSLL h=1	8	0.4147	0.5927	-0.6209	0.4146	0.4145
BSLL h=3	7	0.4617	0.5537	-0.6305	0.4618	0.4609
SJLS h=1	17	0.2553	0.3472	-0.4823	0.2554	0.2559

Figure 5 summarises the prediction and geometry-ablation pattern across the three formal case runs. The left panel reports MAE values directly, while the right panel plots paired MAE differences relative to the full model so that the small SJLS geometry-related difference is visible without exaggerating its magnitude.

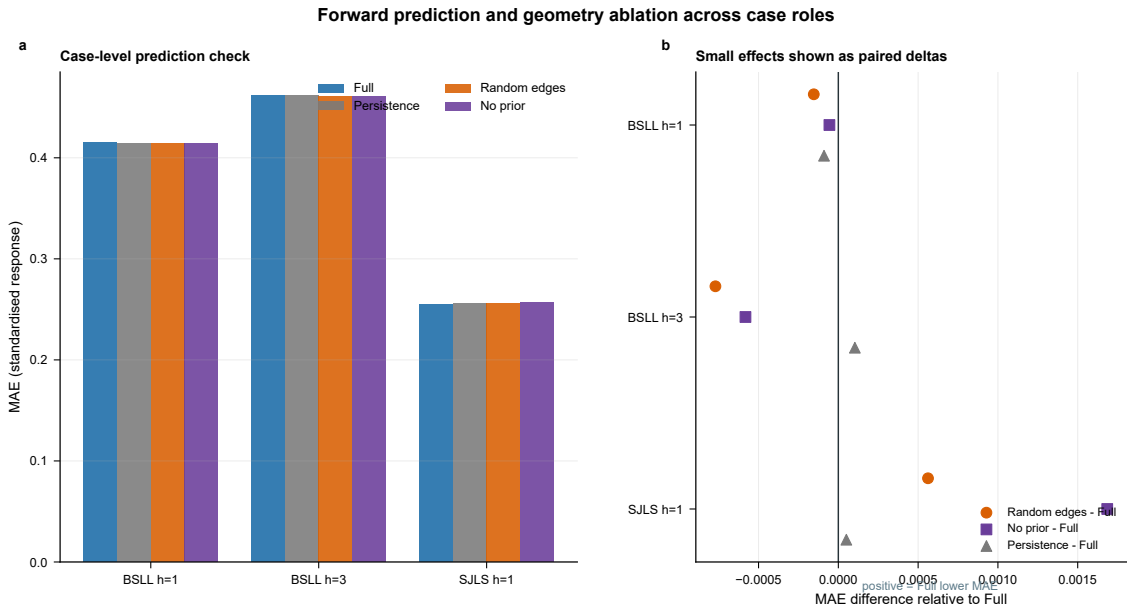


Figure 5: Forward prediction and geometry-ablation summary across the formal case runs. BSLL remains comparable to persistence and random-edge baselines, whereas SJLS provides the clearest case-specific geometry-related difference for the full geometry-aware model. Positive paired deltas indicate variants with higher MAE than the full model.

5.2 Geometry ablation and response consistency

Graph ablations and the role of geometry. The ablations separate two questions that should not be conflated: whether the graph-sequence representation improves prediction, and whether geometry-constrained relations produce interpretable spatial diagnostics. Prediction evidence for the first question is mixed. In SJLS, removing the geometric prior increases MAE from 0.2553 to 0.2570, and randomising rock-machine edges increases MAE to 0.2559. This is consistent with a small case-specific geometry-related difference in the external TSP case. In BSLL, however, randomised edges are comparable to or slightly better than the full model. The BSLL ablation therefore should not be used to claim that constrained topology improves forecasting accuracy.

Figure 6 expands the SJLS result because this is the case in which the geometry-aware model

Table 3: Geometry-related ablation summary. The SJLS case shows the clearest small numerical difference for the full geometry-aware model, whereas BSLL mainly supports the availability of graph-based interpretation rather than a robust predictive improvement.

Case	Full MAE	Random-edge MAE	No-prior MAE	No-monitoring MAE
BSLL h=1	0.4147	0.4145	0.4146	0.4818
BSLL h=3	0.4617	0.4609	0.4611	0.4961
SJLS h=1	0.2553	0.2559	0.2570	1.2500

is most favourable. The figure again shows that the numerical difference is small; its value is to locate the difference in the geometry ablation context rather than to claim a large prediction breakthrough.

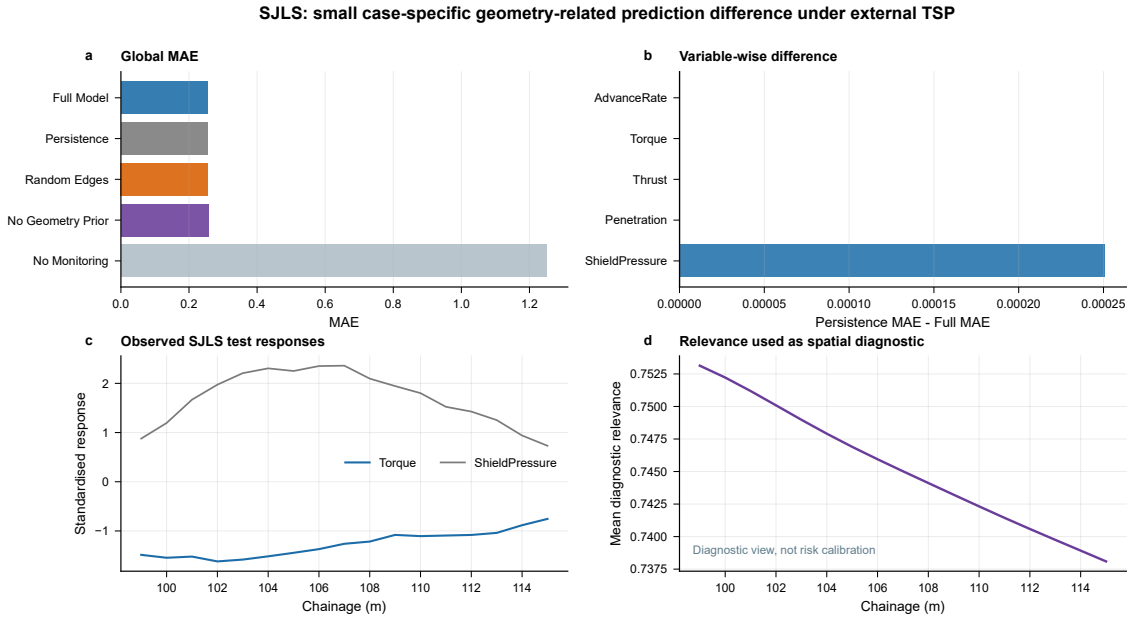


Figure 6: SJLS prediction and geometry-detail view. The full model has a slightly lower global MAE than Persistence, randomised rock-machine edges, and the no-prior variant, while the no-monitoring ablation shows that monitoring history remains essential for this case. The observed test responses and mean diagnostic relevance show the chainage interval over which this diagnostic is evaluated. The relevance curve is a model diagnostic and is not a calibrated risk signal.

5.3 Spatial organisation of learned diagnostic relevance

Because the prediction differences are small, the following sections focus on whether the graph representation produces spatially organised diagnostic relevance.

The interpretation evidence is more consistent across cases. Table 4 reports four post-hoc indicators computed from the saved checkpoints. Diagnostic relevance is positively spatially autocorrelated in all three runs: Moran’s $I = 0.546$ for BSLL h=1, 0.302 for BSLL h=3, and 0.300 for SJLS. This indicates spatially organised relevance fields rather than random surface noise. The degree-control correlations are negative in all cases (-0.229 , -0.324 , and -0.456), which argues against a trivial explanation in which nodes appear important only because they have more incident candidate edges.

The geology association differs by case. BSLL shows a high absolute correlation between mean diagnostic relevance and local TSP velocity ($r = -0.989$ for $h=1$ and $r = -0.688$ for $h=3$), whereas SJLS shows a high positive correlation ($r = 0.870$). The signs should not be interpreted as universal physical rules; they indicate that learned relevance co-varies with the local TSP field within each tunnel setting. The response variable with the highest absolute association also differs: AdvanceRate for BSLL and Torque for SJLS. This case-specific behaviour shows that the diagnostic aggregation route can expose different response-relevance patterns rather than producing a single generic template.

Table 4: Post-hoc spatial diagnostic evidence. Rel.-Geo. r is the Pearson correlation between mean diagnostic relevance and TSP V_p across test chainages. Degree r correlates diagnostic relevance with incident candidate-edge degree; negative values indicate that relevance is not a direct positive proxy for edge count.

Case	Moran's I	Geometry-only I	CV	Degree r	Rel.-Geo. r
BSLL $h=1$	0.546	-0.054	0.018	-0.229	-0.989
BSLL $h=3$	0.302	-0.054	0.005	-0.324	-0.688
SJLS $h=1$	0.300	0.196	0.014	-0.456	0.870

The component CV values are small (0.005–0.018), so the component evidence is best described as component-resolved diagnostics rather than large component separation. Their role is to show that the relevance can be summarised by TBM component without claiming large separation among components.

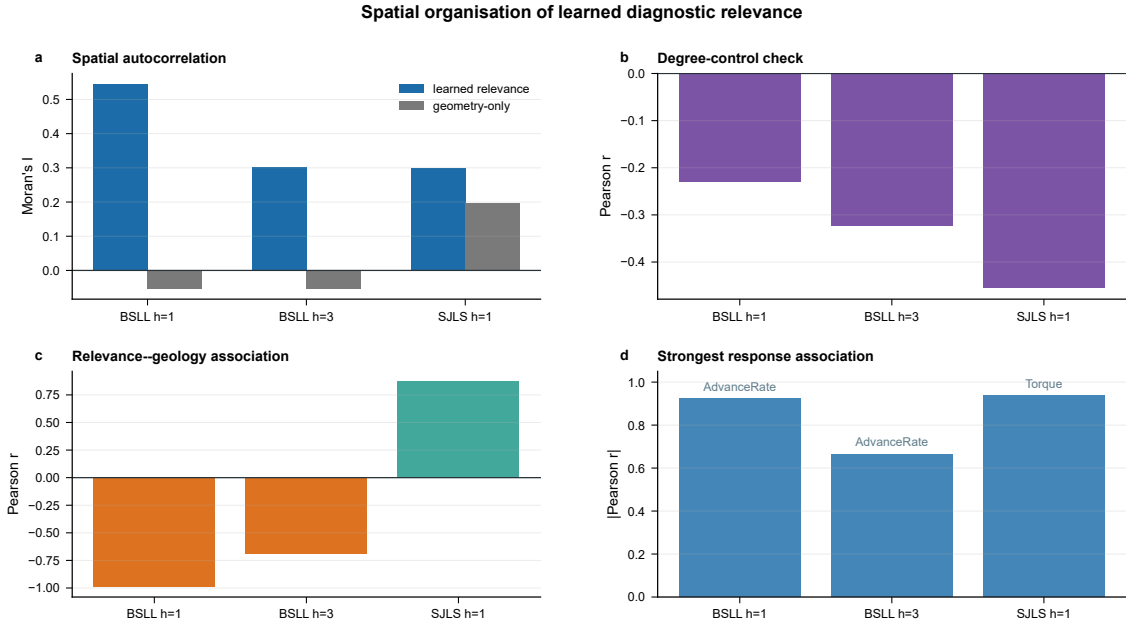


Figure 7: Spatial evidence for learned diagnostic relevance. Learned relevance is positively spatially autocorrelated in all three runs, differs from geometry-only reference patterns, and is not a positive proxy for node degree. The relevance-geology and relevance-response associations are case-specific diagnostics rather than universal physical laws.

5.4 Component-resolved diagnostic relevance along chainage

Figure 8 converts the learned surface-node values into component-by-chainage diagnostic relevance views for the BSLL multi-step case and the SJLS external TSP case. The figure is therefore

a spatial organisation view, not another prediction-performance plot. Figure 9 then focuses on BSLL $h=3$ and aligns mean diagnostic relevance, component shares, and selected monitoring responses on the same chainage axis. Together, these outputs show what the graph-sequence representation adds beyond scalar response prediction: it organises model-derived relevance by TBM surface components and screened rock-machine candidate relations.

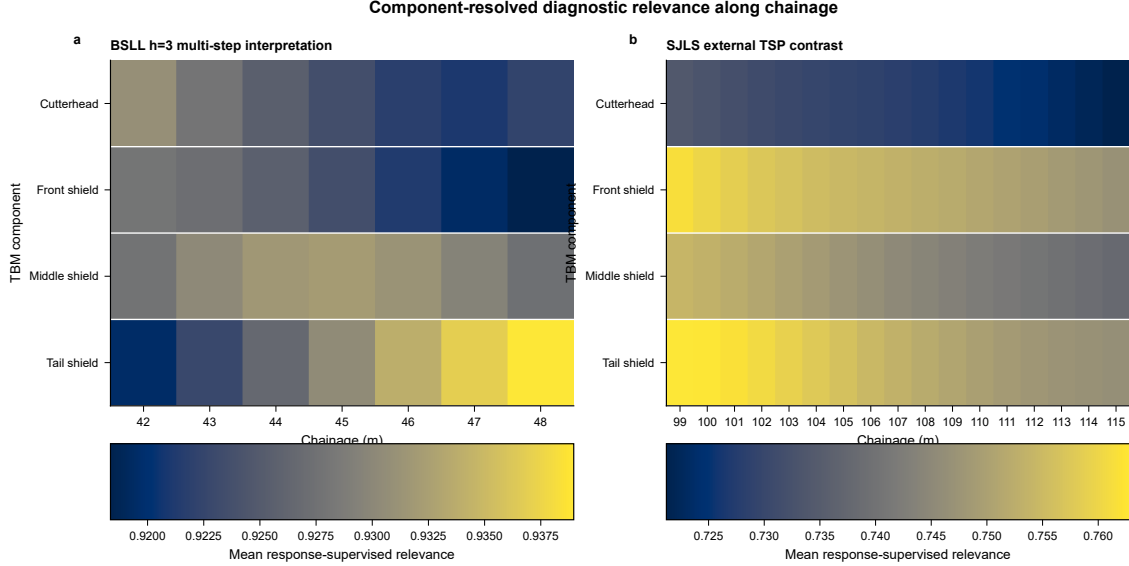


Figure 8: Component-resolved diagnostic relevance aggregated along chainage. The BSLL $h=3$ panel illustrates the multi-step interpretation case, and the SJLS panel provides an external TSP contrast. Colour intensity indicates degree-normalised response-supervised relevance; it is not measured contact pressure or force.

5.5 Robustness and remaining checks

The current evidence supports three bounded claims: the framework remains competitive with temporal baselines, SJLS is consistent with a small case-specific geometry-related accuracy difference, and all three saved checkpoints produce spatially organised diagnostic relevance. It does not support claims of broad global accuracy improvement, measured contact-condition recovery, or operational risk calibration. As a minimal sanity check, paired permutation tests using the saved predictions show that the full model is not separable from Persistence in MAE for BSLL $h=1$ ($p = 0.3948$), BSLL $h=3$ ($p = 0.1659$), or SJLS $h=1$ ($p = 0.3198$), although SJLS has lower MAE than the weaker learned baselines in this run. Multi-seed stability, label-shuffle re-training, threshold sensitivity, spatial detrending, and additional cross-project validation remain necessary before broader robustness claims can be made.

6 Discussion

6.1 What the graph representation adds beyond monitoring-only prediction

The revised experiments sharpen the paper’s central claim. The framework should not be presented as a broad forecasting model. In BSLL, the one-step and three-step runs are essentially tied with Persistence and do not show a consistent benefit from constrained topology. In SJLS, the full model gives a small MAE decrease relative to Persistence, randomised edges, and the no-prior variant, but the difference is modest. This pattern is meaningful because it separates

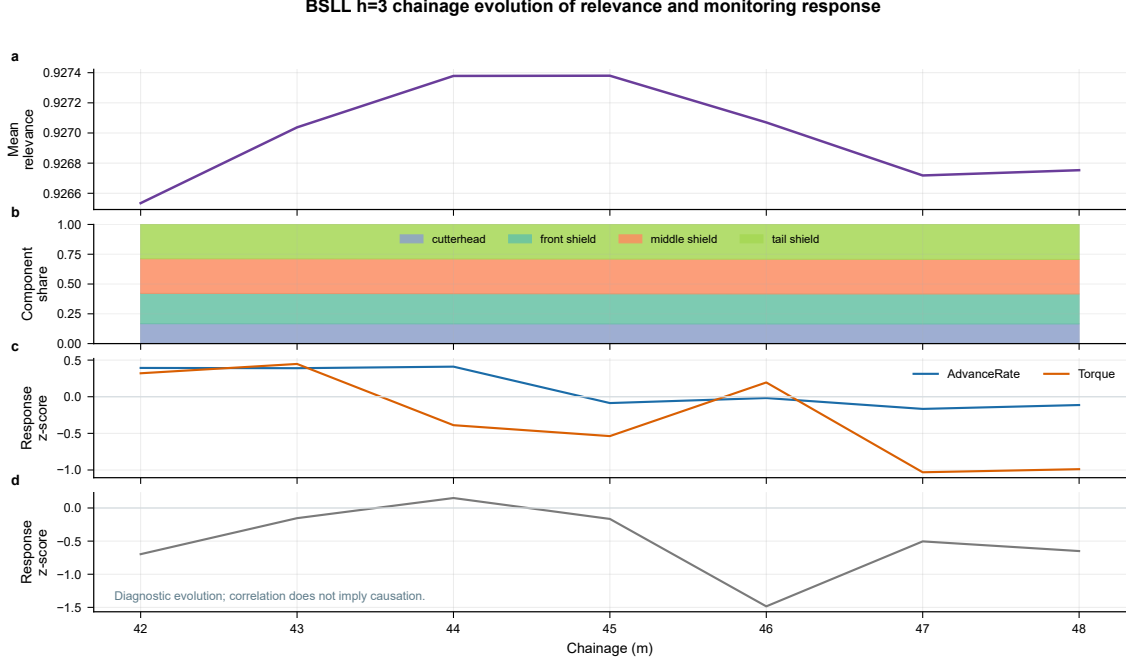


Figure 9: BSL h=3 chainage evolution of diagnostic relevance and monitoring response. Mean diagnostic relevance, component relevance shares, and selected monitoring responses are aligned on the same chainage axis to show how the model diagnostic evolves through the compact future interval. Correlation in this view does not imply causation.

prediction accuracy from representation: the principal value is not forecasting error reduction, but the conversion of a monitoring sequence into an inspectable spatial entity-relation process.

This distinction changes the interpretation of the contribution. The prediction task is used as response supervision and as a consistency check. It is not the only, or even the most consistent, evidence for the method. The more consistent evidence is that TSP voxels, TBM surface components, candidate rock-machine relations, monitoring responses, and diagnostic relevance summaries remain connected in a single spatial representation. The paper’s value therefore lies in making rock-TBM interaction inspectable as a spatial process under incomplete observability. Monitoring-only baselines can predict response variables, but they cannot organise TSP voxels, TBM surface components, and screened candidate rock-machine relations into the same spatial diagnostic structure.

BSLL and SJLS should be read as complementary evidence rather than as a development-validation pair. BSLL provides a compact case in which the limits of prediction are visible: the model is competitive but does not improve on simple temporal persistence, and randomised edges are not worse. This is useful because it prevents the paper from overclaiming geometry as an accuracy mechanism. At the same time, BSLL provides interpretable multi-step and component-chainage examples that illustrate what a graph-sequence representation adds beyond scalar prediction.

SJLS plays a different role. It uses an external TSP-derived velocity field and a larger number of forward test samples. In this case, the full geometry-aware model has slightly lower MAE than the ablated variants, and the learned relevance has a high absolute association with the local TSP field. This makes SJLS the clearer case for arguing that geometry-constrained representation can support response-consistent prediction and spatial interpretation. The important point is not that SJLS proves a large prediction improvement, but that it shows the same representation operating in a second tunnel setting with a clearer geometry-ablation pattern.

6.2 Why prediction gains are limited

The prediction results do not show a broad accuracy improvement, and several factors explain why this is expected rather than surprising. First, the test sets are small, especially for BSLL, which limits the statistical power of any accuracy comparison. Second, the mileage-ordered split is a forward extrapolation setting: validation and testing are performed on later continuous tunnel intervals that may exhibit geological distribution shift, so Persistence is a strong baseline that benefits from local autocorrelation. Third, TBM monitoring responses are themselves strongly time-autocorrelated, which makes short-horizon forecasting difficult to beat with any added spatial structure. Fourth, TSP-derived geological attributes and TBM monitoring responses operate at different spatial and temporal scales, and the current alignment may not fully exploit the geological information. These factors together explain why the full graph-sequence model remains comparable to Persistence in BSLL and shows only a small geometry-related difference in SJLS.

This limited prediction gain does not negate the contribution. The graph representation provides spatially organised rock-machine relation diagnostics that temporal baselines cannot produce. Moran’s I is positive in all three runs, and the degree-control correlations are negative, supporting a bounded but useful claim: the diagnostic relevance views are not spatially random and are not simply inflated by the number of candidate edges incident to a TBM node. The experiments motivate a GIScience framing of TBM excavation analysis: the problem is not merely to add more features to a forecasting model, but to formalise heterogeneous spatial entities with different supports and aggregate model-derived relevance into spatial views that support inspection, comparison, and case-based reasoning.

6.3 Interpretation boundary and future validation

Several limitations remain. The test sets are small, especially for BSLL, and all results are based on single fixed-seed formal runs. Negative R^2 values show that forward extrapolation remains difficult under geological shift. The observed attention-geology correlations may partly reflect chainage trends and therefore require spatial detrending before broader geological interpretation is made.

Future work should prioritise multi-seed stability, label-shuffle controls, candidate-edge threshold sensitivity, spatial detrending, and additional tunnel sections. Independent event labels, expert assessments, contact measurements, or shield-pressure distribution observations would be required before the relevance maps could be connected to operational risk or actual contact conditions. Until then, the defensible claim is representation and response-consistent interpretation, not contact-condition recovery or general predictive dominance.

7 Conclusion

This study proposed a response-supervised, geometry-constrained graph sequence framework for spatially explicit rock-TBM interaction representation. The method formalises TSP-derived rock voxels and TBM surface components as heterogeneous spatial entities, screens candidate rock-machine relations by geometric compatibility, and aggregates learned relevance into component- and chainage-resolved diagnostic views.

The revised experiments clarify the strength and boundary of the contribution. Across BSLL DyK1017+205 and SJLS Dyk1252+411, the framework remains competitive with temporal baselines but does not establish a broad, statistically detectable accuracy improvement. BSLL mainly supports the multi-step and diagnostic-interpretation use case, while SJLS provides the clearest small case-specific geometry-related prediction difference. The consistent evidence is interpretive: all three saved checkpoints produce spatially organised diagnostic relevance, and the degree-control checks indicate that relevance is not simply a proxy for candidate-edge count.

The main contribution is therefore not a claim of broad forecasting improvement but a spatial entity-relation representation and an auditable route from response supervision to component- and chainage-resolved diagnostic relevance. Future work should test multi-seed stability, label-shuffle controls, threshold sensitivity, spatially detrended geology association, and additional tunnel cases before broader claims are made about contact-condition recovery or operational risk estimation.

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Data availability statement

The raw TBM monitoring and TSP data are project data and are not publicly available because they contain engineering and site information. The processed experiment configurations, analysis scripts, and derived result summaries used to reproduce the reported tables and figures are available from the corresponding author upon reasonable request, subject to project data-use permissions.

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